Dathan agreed. And, predictably, a few of these configurations using a liquid engine showed satisfactory results in simulations. In the next group discussion, when some of the analyses were presented, and the results turned out to be as usual unsuccessful, I remarked, ‘Dathan, why don’t you show us the results of that liquid-inclusive configuration.’

Dathan knew he was trapped. Hesitantly, he presented the simulation results, which showed satisfactory performance of a PSLV using a liquid engine as the second stage. But my detractors wouldn’t budge.

‘Nobody questions the efficiency of a liquid engine,’ said one. ‘But we are not considering it because we don’t have such an engine and we can’t develop one.’

It was time to do something drastic.

I went to PSLV planning group head Srinivasan’s house. He was surprised, since there was nothing we couldn’t discuss in office. But I chose to corner him in his den, where he would be free to react the way he wanted to, without official civility. After all, I was going to be harsh.

After turning down a coffee, I told him, ‘Srinivasan, you are at the cusp of making history for India. You have two options: one, to be remembered as the person who headed PSLV, the most successful rocket India has made. Two, as the person who destroyed Vikram Sarabhai’s dream of making India a space power.’

As Srinivasan sat stunned, I continued, ‘You know you can’t launch a PSLV without a liquid engine, but you are scared to oppose the chorus against it. I promise you to deliver a workable liquid propulsion engine. Think about it and allow a discussion on liquid engine at the next meeting.’ I walked out.

In the coming days, I frequented VSSC director Brahm Prakash’s office to impress upon him the need to ground-test the Vikas engine we had developed. Brahm Prakash, amidst a bunch of squabbling scientists, walked tall and commanded respect from everyone. Everyone knew that Brahm Prakash, a metallurgist from the Masssachusetts Institute of Technology who headed the metallurgy department at IISc, had agreed to take over as the first director of VSSC in 1972 after much pleading by Dhawan himself.